Avi-Defender

SMART BIRD MONITORING & DETERRENT SYSTEM

presented by:

Min, Habib,
Ryan and Louis

Skit Time!

💡 "Bird intrusions are a persistent issue, but what if we could stop them intelligently and automatically?"

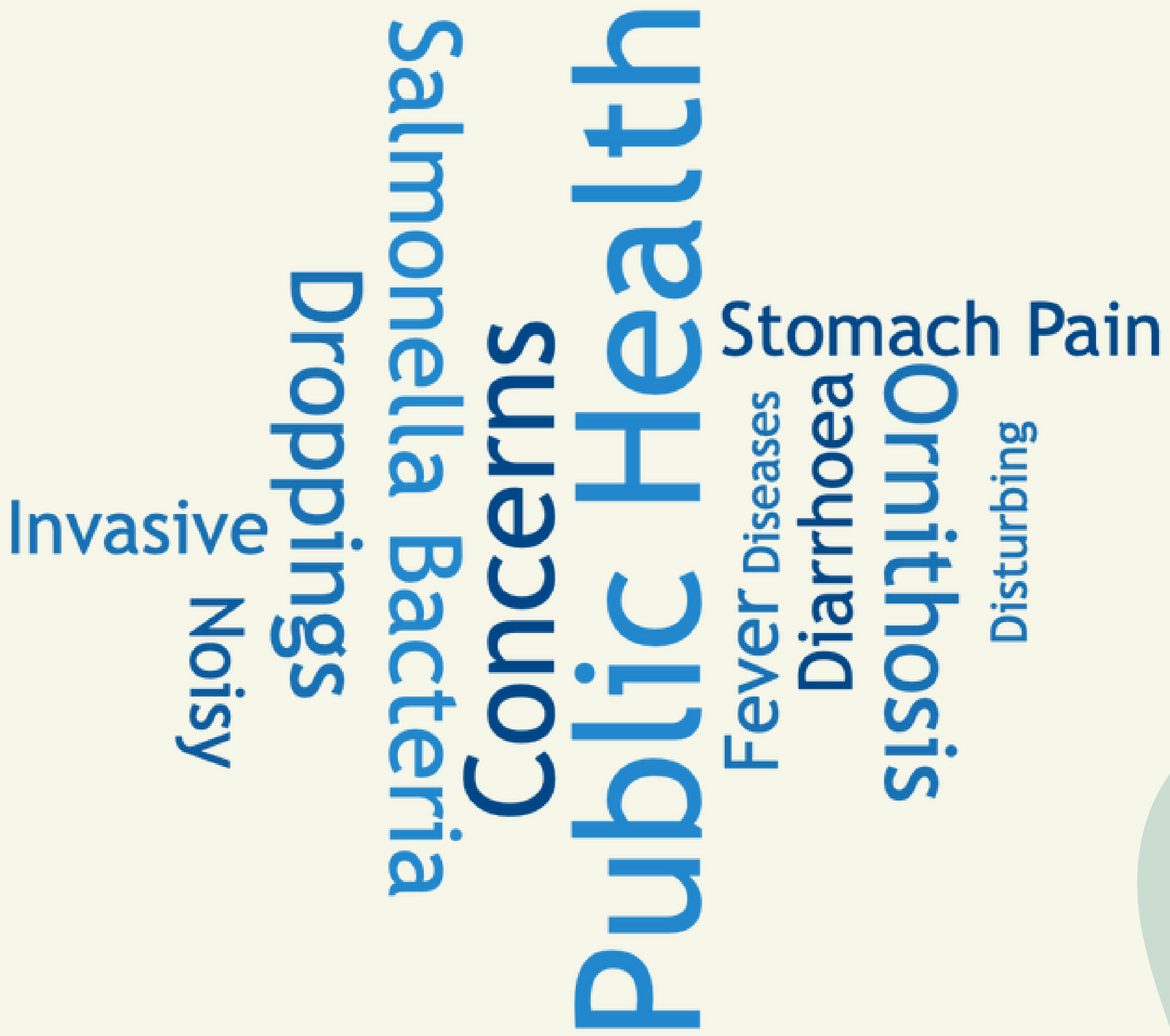
Skit Time!

🎭 Setup:

- Imagine you're returning a tray at a hawker center.
- You see birds swooping in, scavenging leftover food.
- Messy, unhygienic, and unappealing for diners!

💡 "Bird intrusions are a persistent issue, but what if we could stop them intelligently and automatically?"

The Problem - Why Birds Are a Concern?



Problem? 🥡 Birds scavenge for food → Contamination risk!

Ref: CNA Explains: Singapore has a pigeon problem. What's being done about it?

AviDefender

INTRODUCING Avi-Defender

📢 An **IoT-based Bird Detection & Deterrent System** for maintaining hygiene in public spaces.

💪 **Automates** bird detection and deterrence using **ultrasonic** sensors and **cameras**.

⏳ Provides **real-time** alerts to help staff manage bird activity efficiently.

✅ **AI-powered** detection & deterrence ✅ **Scalable** for multiple locations

AviDefender

POTENTIAL LOCATIONS, USERS & BENEFICIARIES

Location: Hawker Centres & Food Courts
Users/Beneficiaries: Stall owners/ Diners

Location: Outdoor Restaurants & Theme Parks
Users/Beneficiaries: Stallowners/Diners/Visitors

Location: Farms
Users/Beneficiaries: Farm Owners

🚀 Hands-Free Bird Control – Automates deterrence, freeing up staff.
🧼 Less Cleanup, More Hygiene – Prevents mess, keeps spaces cleaner.
🍽️ Food-Safe Environments – Stops birds from contaminating open food areas.

AviDefender

PROJECT GOALS/IMPACTS

- **Non-harmful** IoT system 🌿
- Bird detection via **ultrasonic sensor** & **camera** 📡
- **Audio-based** deterrence 🔊
- **Real-time** configuration
- Data logging & insights 🖥️

- 🏡**Hygienic** outdoor spaces
- 🌾 **Fewer** crops loss
- 👨‍💼**Efficient** operations for staff

AviDefender

AVI-DEFENDER - END-TO-END PROCESS

1 **Detect** → Pi Camera + YOLOv11Nano identify birds

2 **Track** → Sense HAT maps bird positions

3 **Verify** → Ultrasonic & PIR sensors confirm presence

4 **Deter** → Buzzer activates if conditions match

5 **Log** → MySQL stores data, Website visualizes trends

AviDefender

RASPBERRY PI FUNCTIONS

🎯 Enhances detection & system management

⚙️ Key Functions:

- 📸 AI bird detection (Camera + YOLOv11n)
- 📊 Logs bird count & location (MySQL)
- 🔄 MQTT communication with Arduino

🛠️ Optimisations:

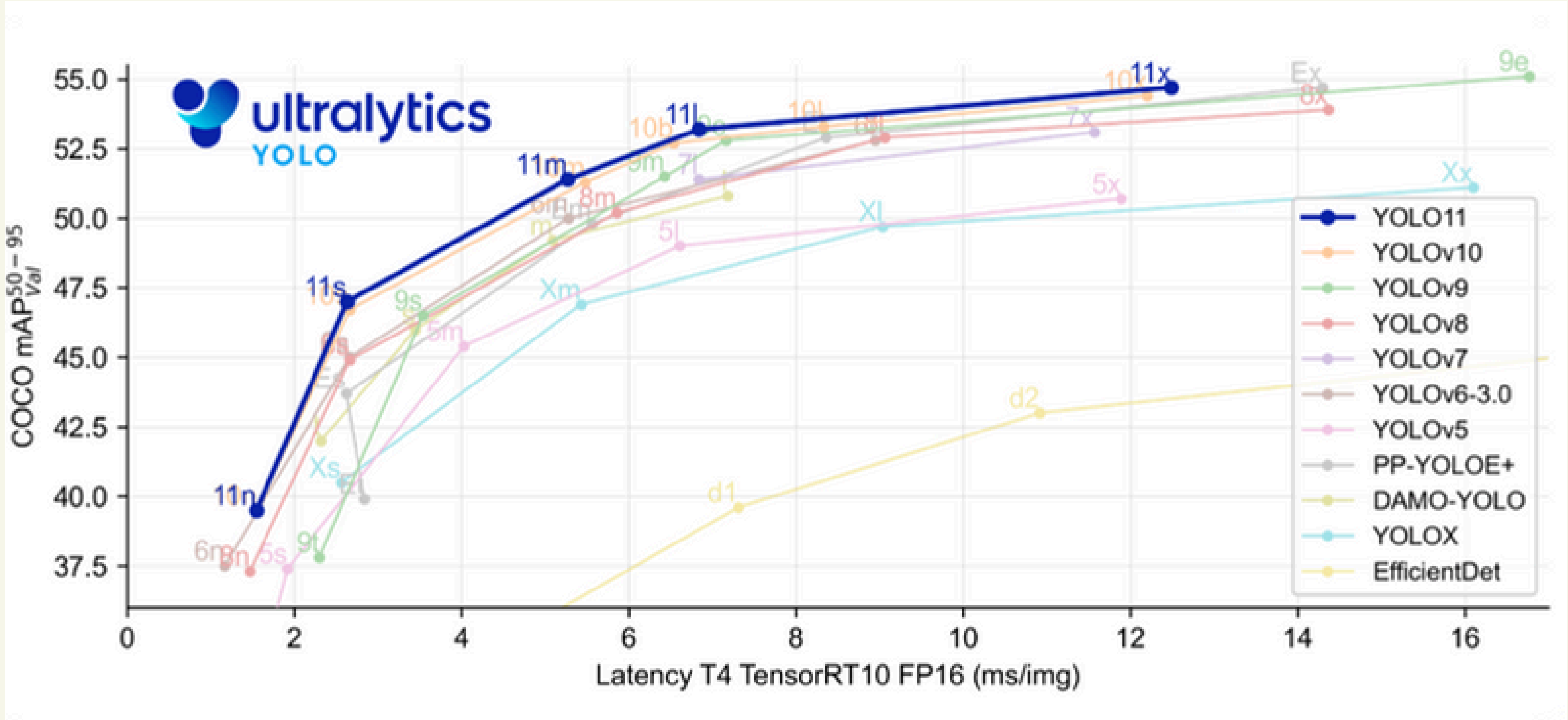
- ✅ Multi-threading for speed
- ✅ TensorFlow Lite for efficiency
- ✅ Simplified live feed (dots & circles)

AviDefender

Our Model : YOLOv11Nano

Why Yolov11Nano?

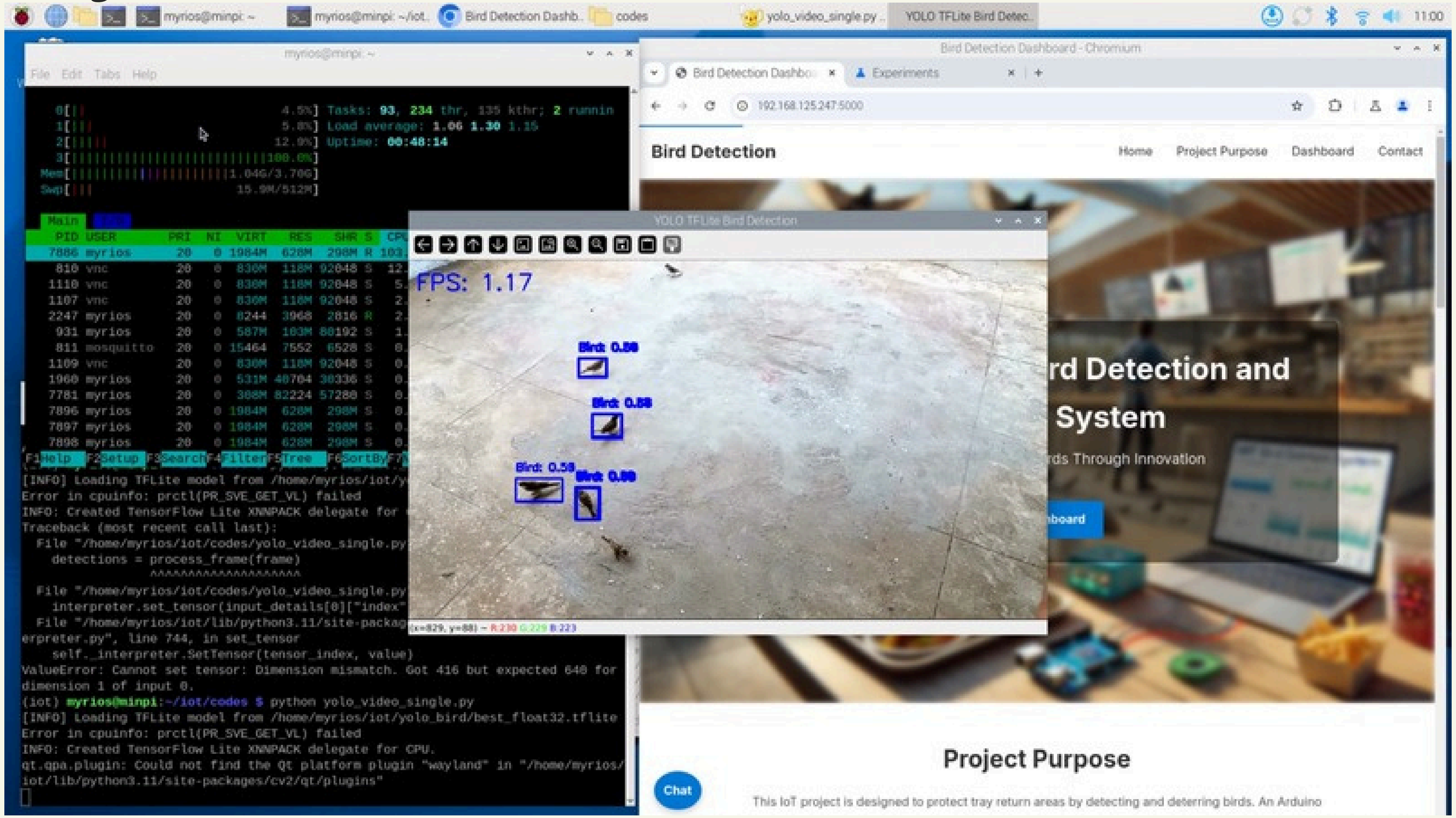
- Lightweight Model Optimized for Edge Devices
- Faster Inference with Lower Computational Cost
- 20 FPS Capable, Limited by Single-Frame Capture
- Trained on 10,000+ Images for Bird Detection



AviDefender

YOLO OPTIMIZATION

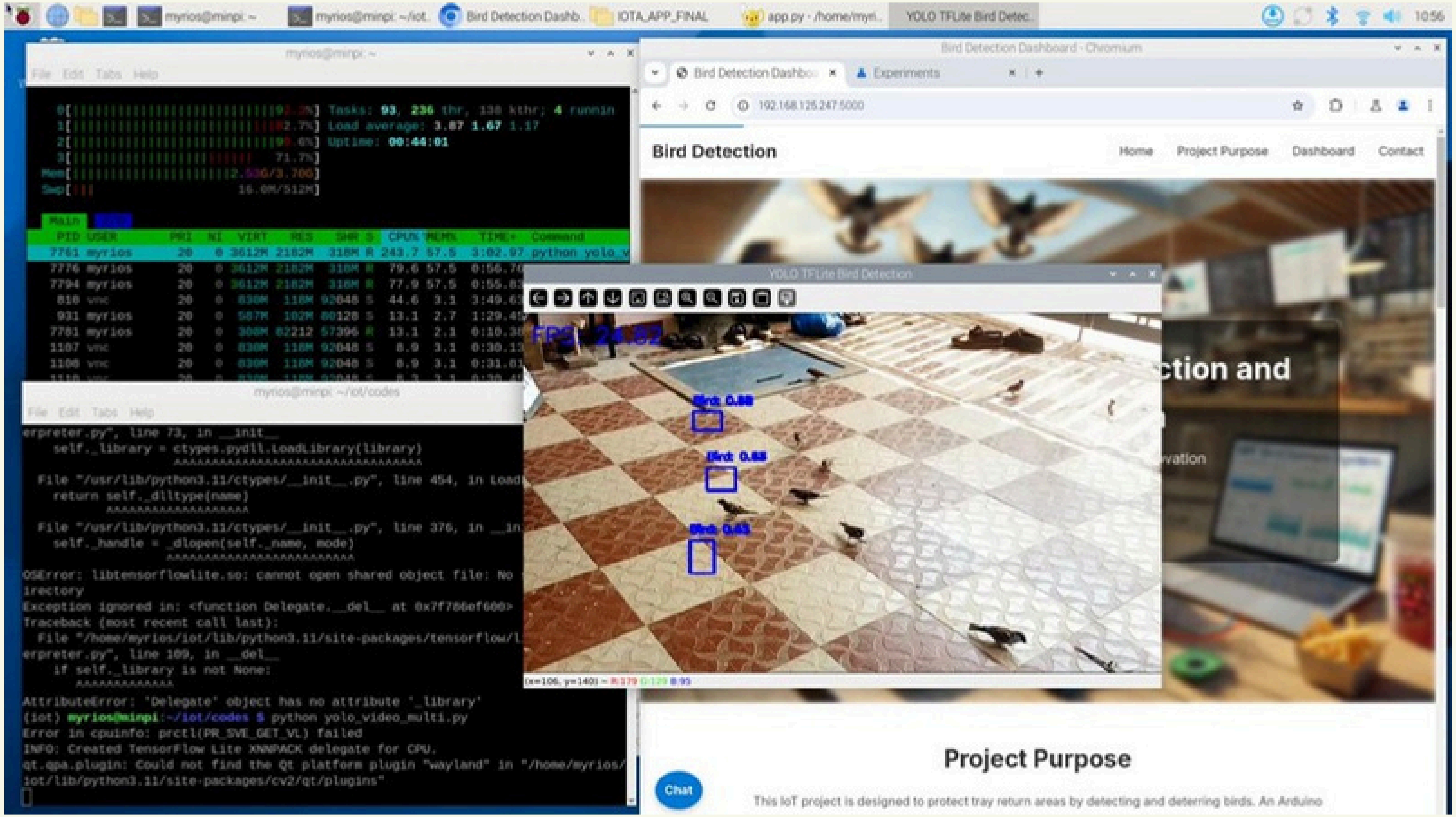
Single-thread, tf-lite



AviDefender

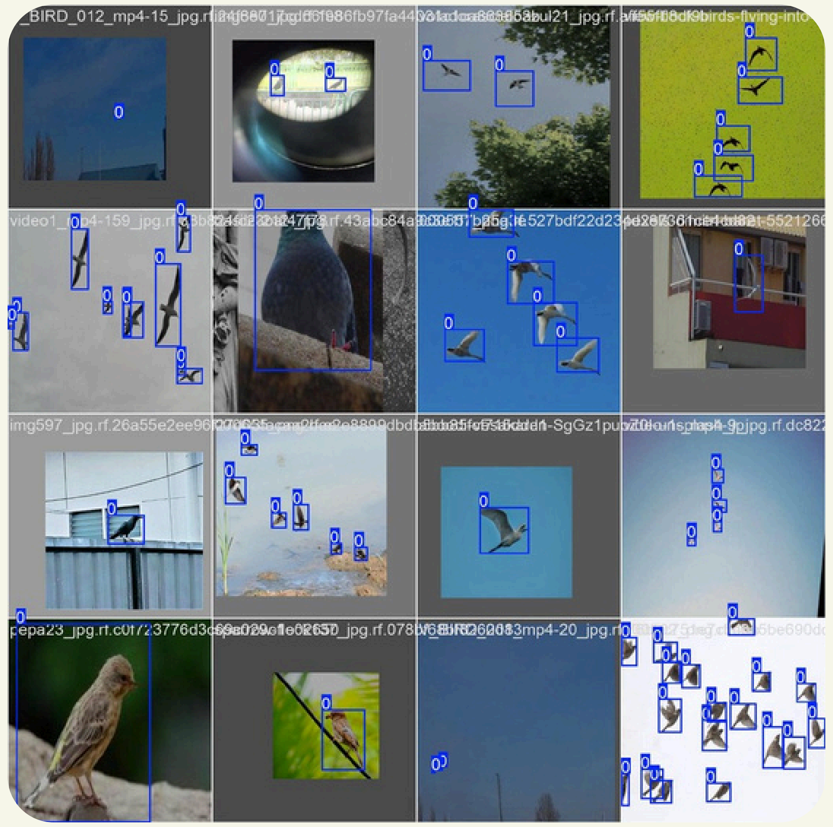
YOLO OPTIMIZATION

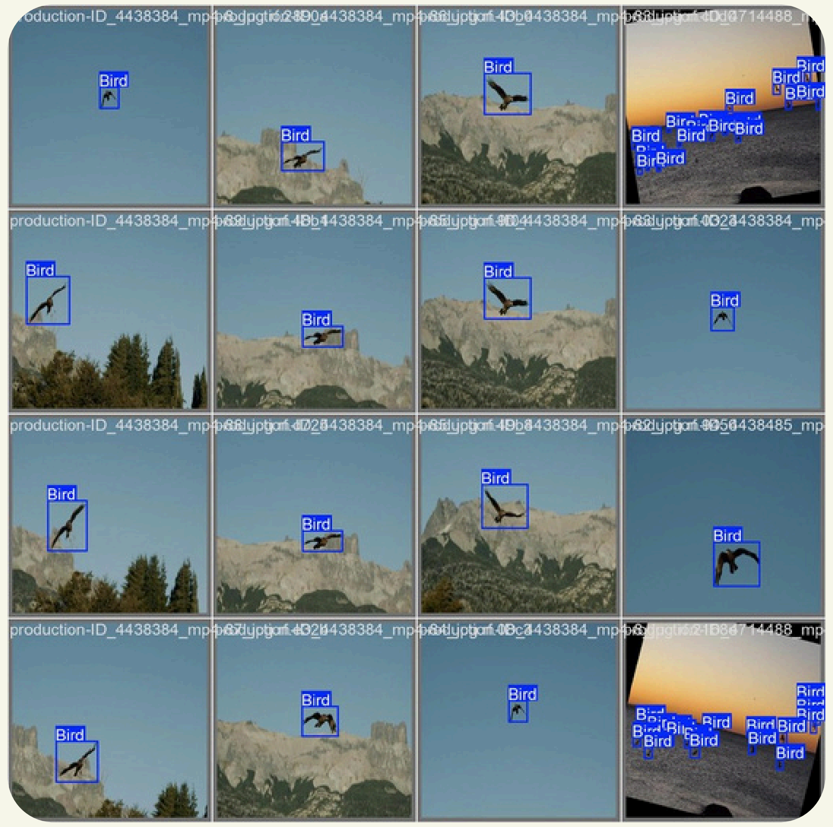
Multi-thread, tf-lite



AviDefender

Output Examples





AviDefender

SENSE HAT & REAL-TIME VISUALIZATION

🟦 Maps Detections → Displays bird locations on an 8x8 LED matrix.

🔵 Dynamic Tracking → Blue pixels show presence, intensity indicates proximity.

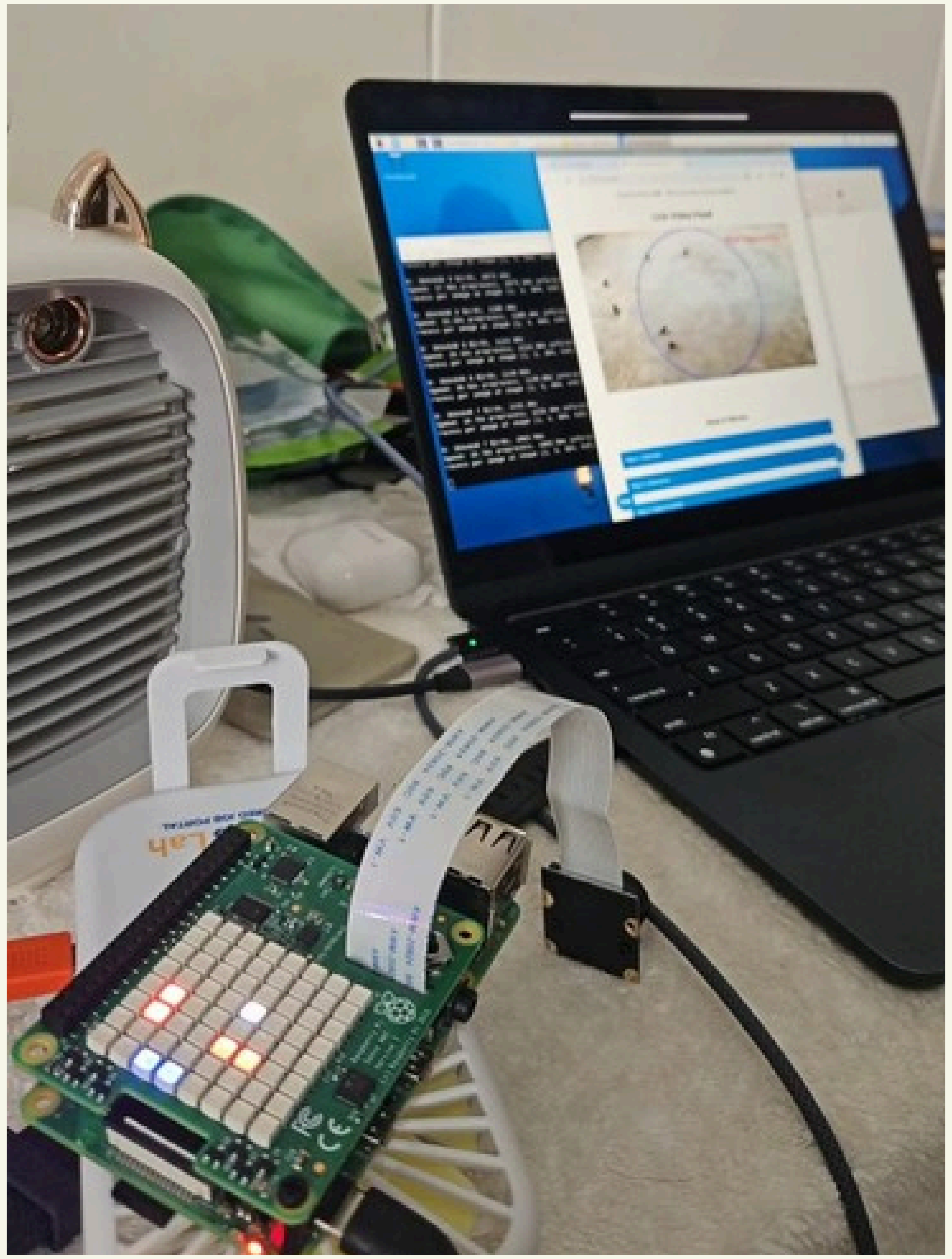
🖼️ Suggested Images:

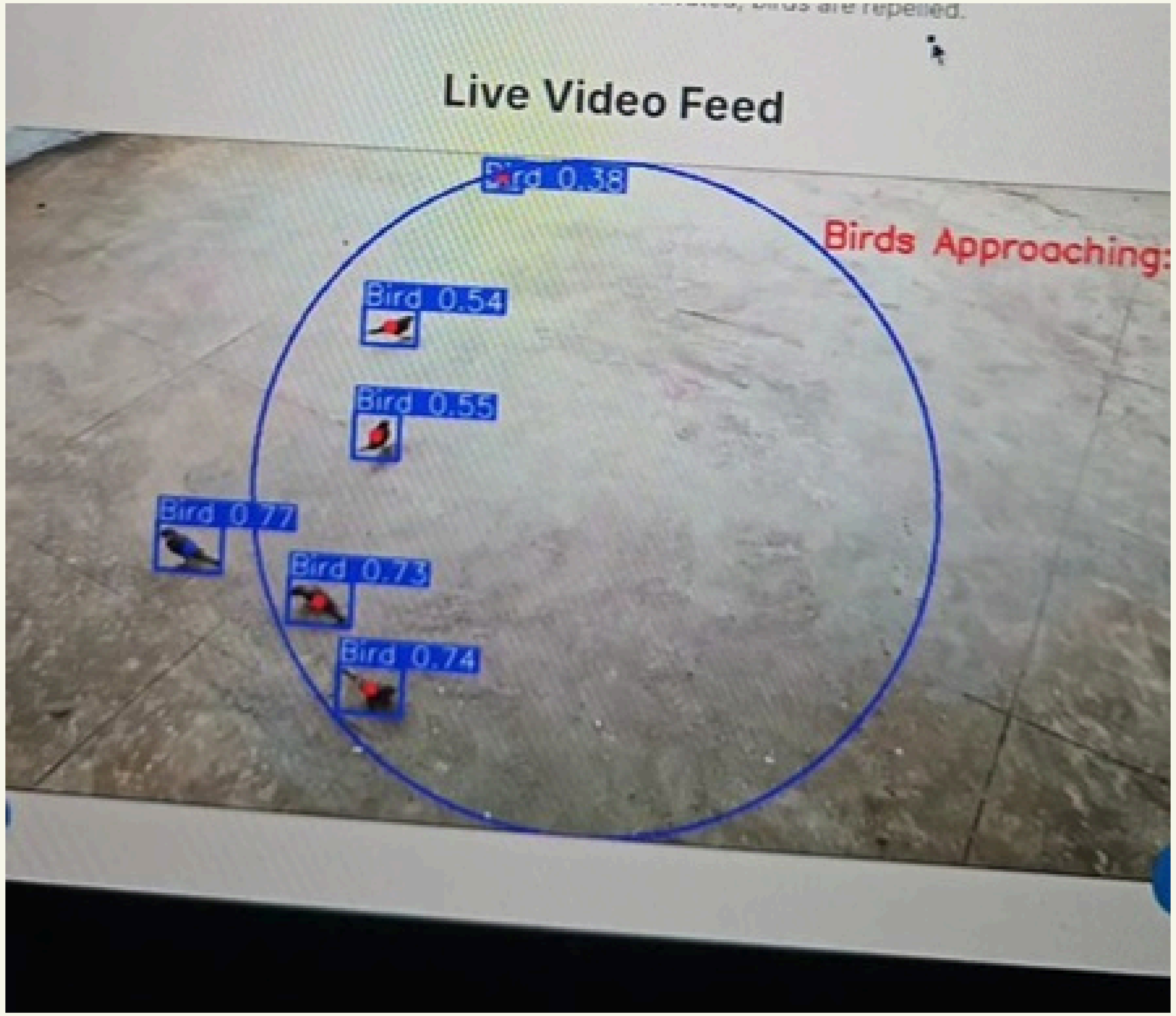
- Diagram showing Raspberry Pi's role in detection & data flow.
- Sense HAT display mapping bird positions

Sensehat

AviDefender

ORIGINAL YOLO
with SenseHat update





AviDefender

REAL-TIME DATA INSIGHTS DASHBOARD FLOW

1️⃣ Bird Detection
🔍 Ultrasonic sensor & camera detect birds.

2️⃣ Data Processing
💻 Raspberry Pi logs bird count & localisation and timestamp in MySQL

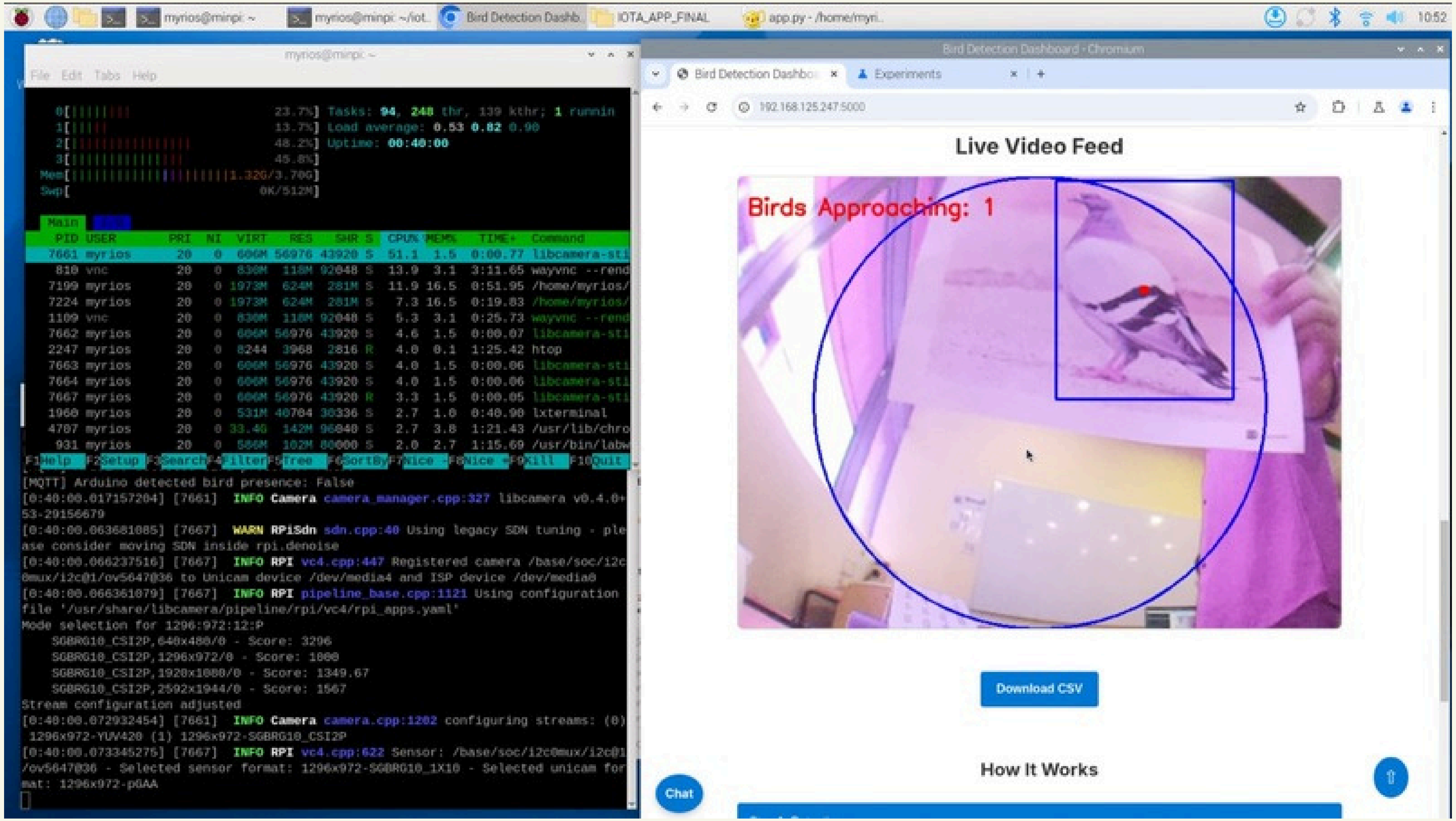
3️⃣ Live Dashboard Updates
📊 Displays real-time bird activity & deterrent actions

4️⃣ Alerts & Insights
🚨 Sends notifications for frequent bird activity & analyses trends

⚙️ Customisable Settings – Adjust detection range & deterrent intensity.
💻 Easy Monitoring – Web dashboard for instant control & data visualisation.

AviDefender

EFFICIENT LIVE VIDEO FEED



Low CPU usage even when YOLO, MQTT, Python Flask & MySQL are running concurrently on the Raspberry Pi 4B

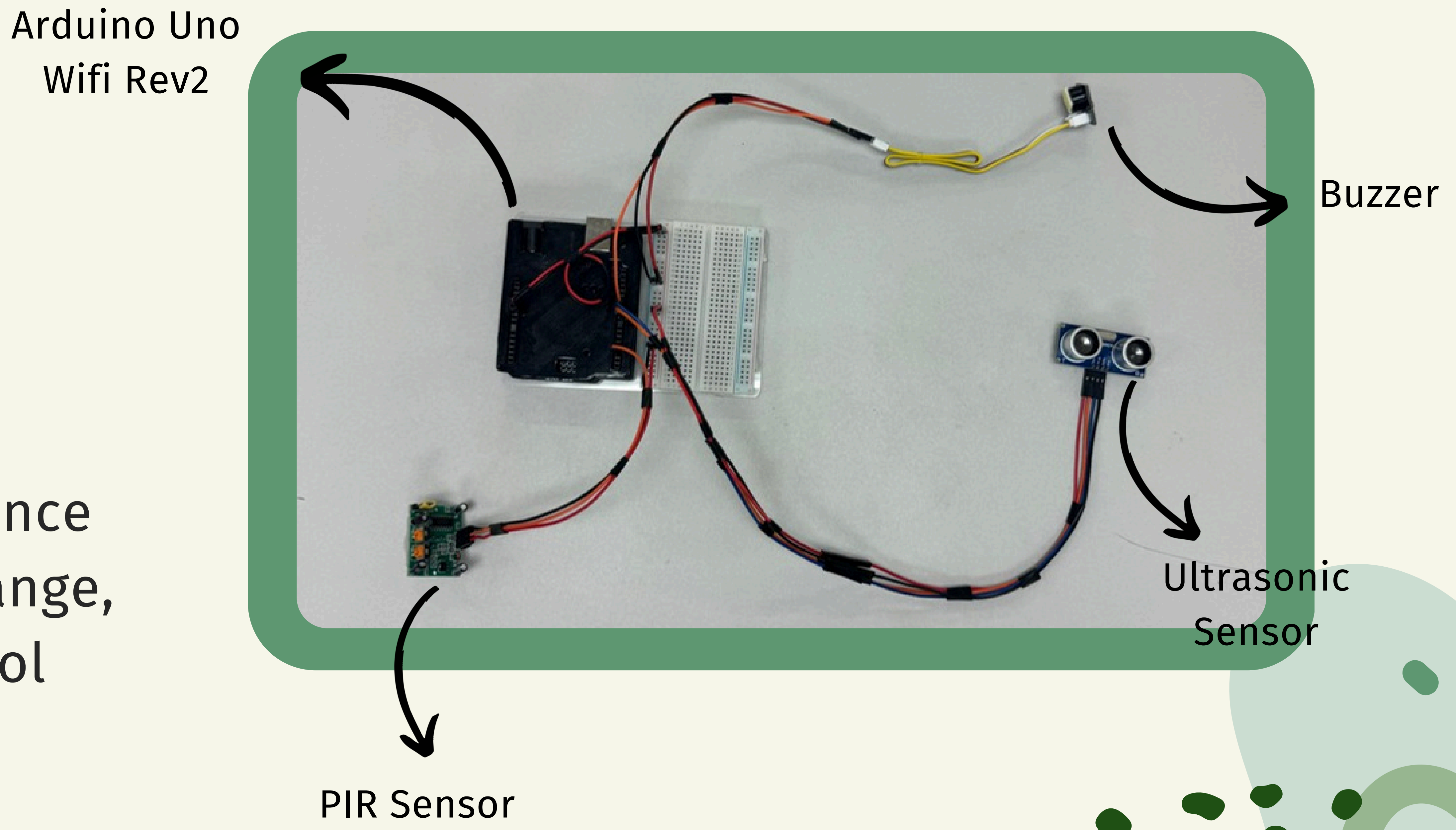
AviDefender

🤖 ARDUINO - DETECTION & DETERRENCE MODULE

🎯 Detects & deters birds using ultrasonic & PIR sensors

⚙️ Key Features:

- 📡 Senses movement & communicates via MQTT
- 🔊 Activates buzzer for deterrence
- 🛠️ Customisable: Adjustable range, sound frequency, on/off control

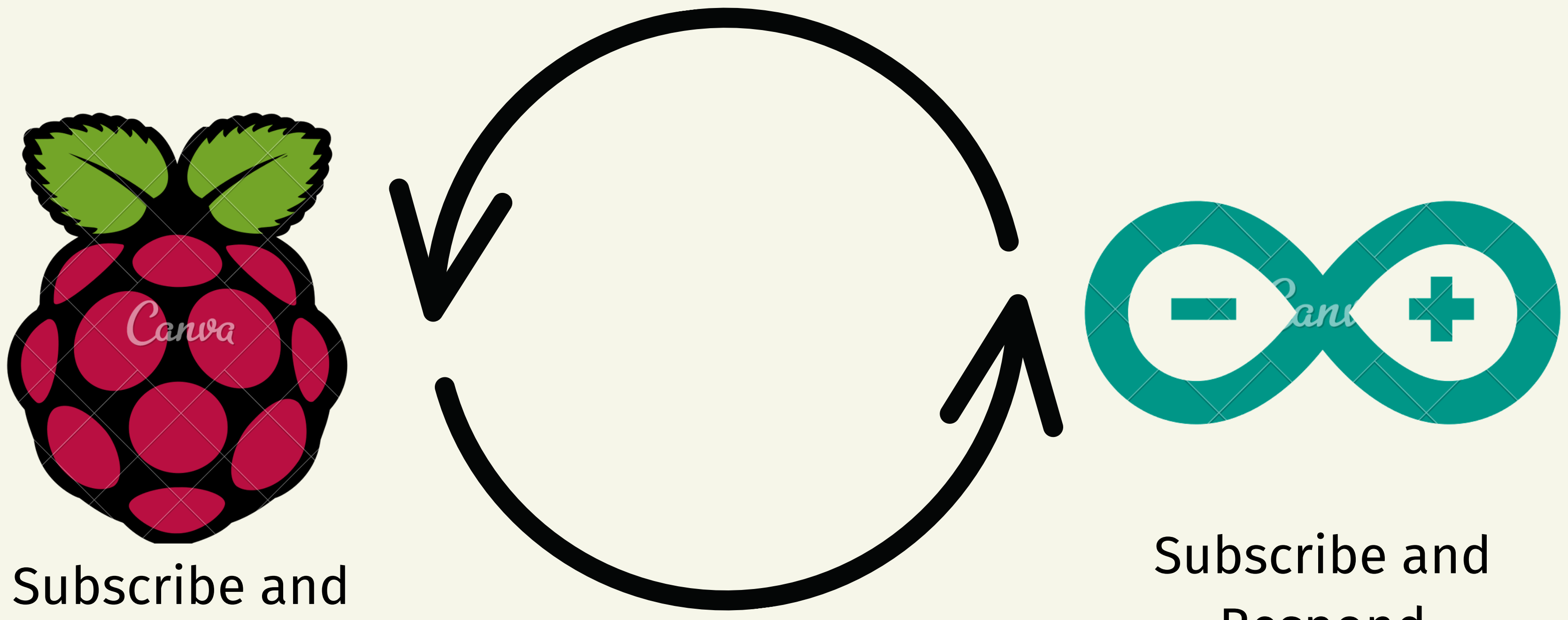


AviDefender

MQTT COMMUNICATION IN AVI-DEFENDER

📡 How MQTT Enables Smart Communication

- Raspberry Pi as MQTT Broker & Publisher
  - Controls all message flow and publishes detection & deterrent commands.
- Arduino as Subscriber & Responder
  - Listens for detection data and activates deterrents when necessary.



Subscribe and Publish

Subscribe and Respond

AviDefender

📡 HOW AVI-DEFENDER COMMUNICATES USING MQTT

🔄 Process Overview: Pi detects bird → Publishes data → Updates settings → Arduino activates buzzer if conditions match → Logs status → Turns off when bird leaves.

Topic	Published By	Subscribed By	Message Type
`bird/detection/presence`	Raspberry Pi	Arduino	"true" or "false" (bird detected or not)
`bird/detection/range`	Raspberry Pi	Arduino	Integer (e.g., 50 for 50 cm detection)
`buzzer/pitch`	Raspberry Pi	Arduino	Integer (e.g., 1500 Hz buzzer frequency)
`bird/detection/deterrent`	Raspberry Pi	Arduino	"true" or "false" (enable/disable deterrent)
`buzzer/activated`	Arduino	Raspberry Pi	"true" if buzzer is on, "false" if off

📊 Table: MQTT Topics & Message Flow

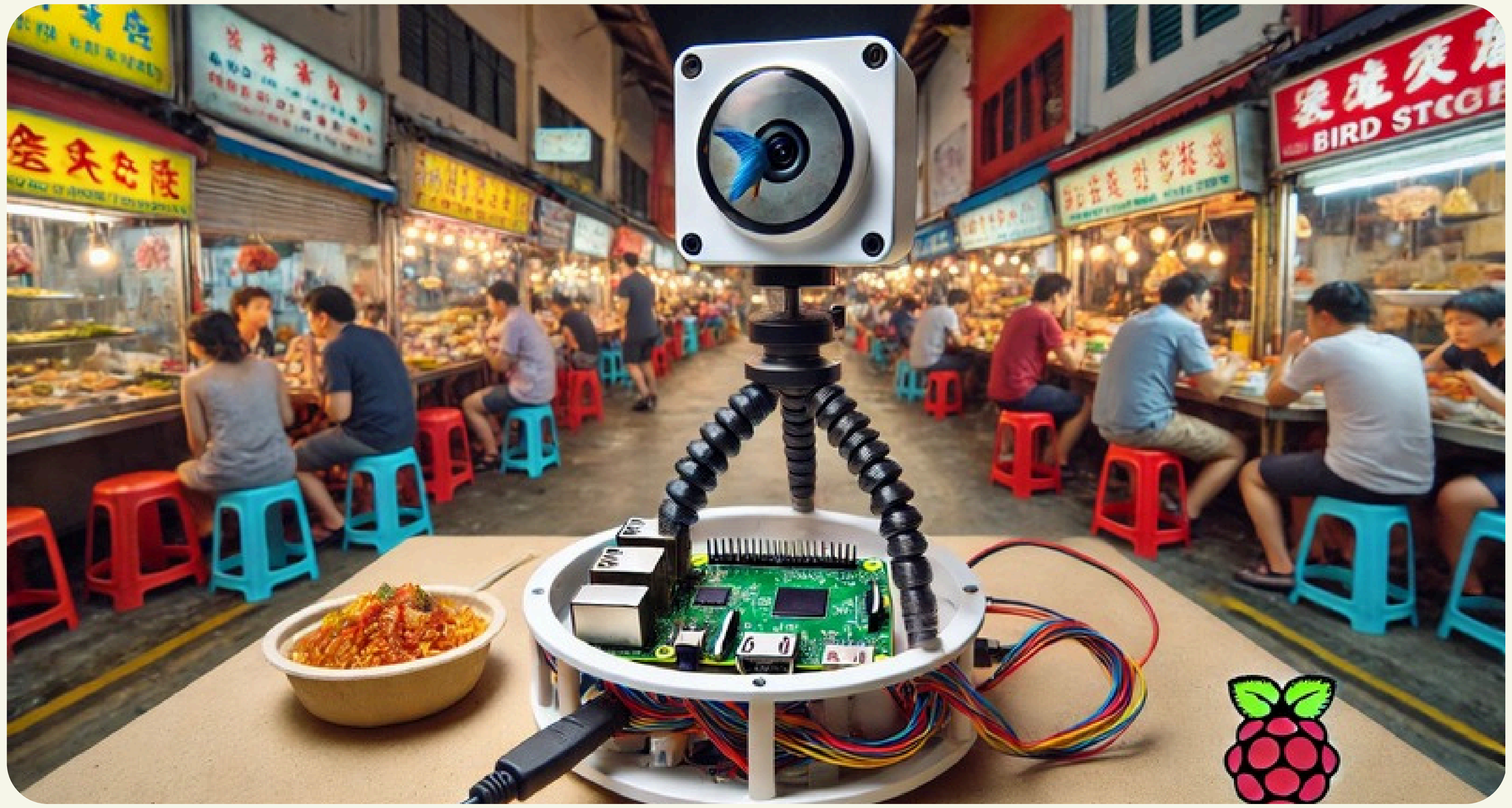
AviDefender

OUR FUTURE IMPROVEMENTS

AviDefender

🎥 1. LIVE VIDEO DETECTION - THE NEXT STEP

- ✅ Upgrading camera for continuous real-time monitoring
- ✅ YOLOv11Nano optimised for high-FPS video input
- ✅ Rotating Pi setup enables smooth video feed



AviDefender

📊 2. USING DATA FOR SMART DECISION-MAKING

- ✅ Analyze historical bird movement patterns
- ✅ Optimize deterrence times for higher effectiveness
- ✅ Machine learning models for predictive analysis

AviDefender

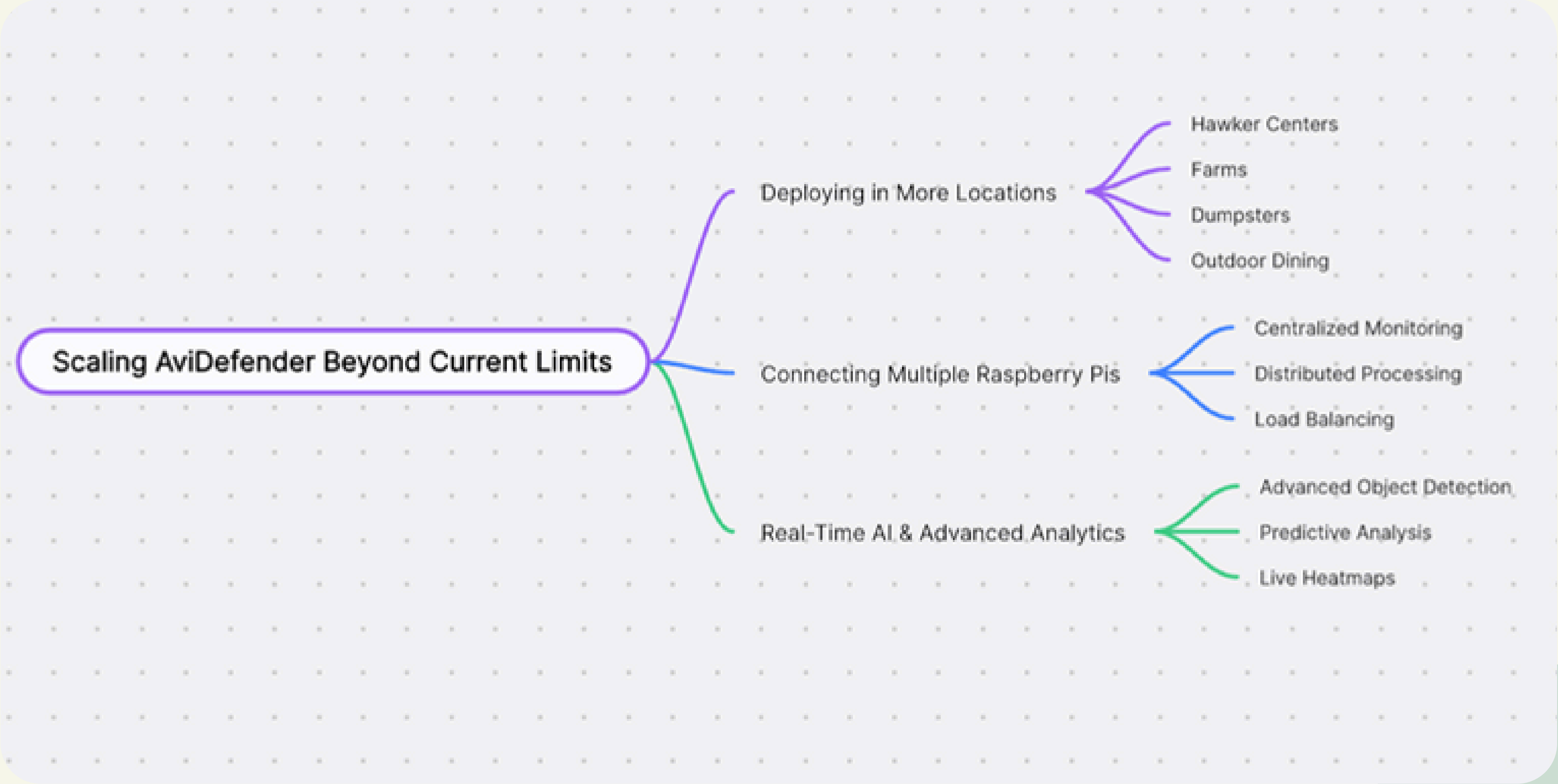
🌦️ 3. ADDING NOISE & WEATHER SENSORS

- ✅ Dynamically adjusts deterrence based on weather conditions (whether birds come more or less on a sunny/rainy day)
- ✅ Noise levels influence bird detection accuracy
- ✅ Enhances adaptability across locations

AviDefender

🚀 4. SCALING AVI-DEFENDER BEYOND CURRENT LIMITS

- ✅ Deploying in more locations
- ✅ Connecting multiple Raspberry Pis for larger monitoring networks
- ✅ Real-time AI & advanced analytics



AviDefender

DEMONSTRATION

AviDefender

💡 WHY AVI-DEFENDER?

- ✅ AI-driven smart deterrence
- ✅ Scalable & cost-effective
- ✅ Future-proof with AI & video analytics

Before

After

AviDefender

DID WE MEET OUR GOALS/IMPACTS?

Hygienic Spaces → Prevents contamination with real-time bird deterrence.

Crop Protection → Detects and scares birds before damage occurs.

Efficient Operations → Reduces manual intervention and cleanup.

AviDefender

CONCLUSION

- ✅ AI & sensors detect & deter birds.
- ✅ Prevents contamination, ensuring hygiene.
- ✅ Real-time alerts & data logging for analysis.
- ✅ Raspberry Pi & Arduino communicate via MQTT.
- ✅ Customizable deterrents (range, sound frequency).
- ✅ Future-ready: Sensors, analytics, scalability.

AviDefender

Q&A

AviDefender

THANK YOU